

KB-006-Slow-Charging

KB-006 — Slow or Throttled Charging

Category: Charging Performance

Call Volume: ~6% of contacts

Resolution Type: Mostly educational; some hardware resets

Primary Remote Actions: Soft reset, power output check

Why This Matters

Slow charging complaints are among the most nuanced in EV support. Most of the time, the charger is performing correctly and the reduced speed has a legitimate technical explanation — battery state of charge, thermal management, power sharing, or a vehicle charge limit. However, a small portion of slow-charging complaints reflect genuine hardware faults, such as a degraded power module or a throttled rectifier. Agents must quickly distinguish “expected behavior” from “actual fault” to avoid unnecessary tech dispatches while also not dismissing drivers experiencing real problems.

Background: Why EV Charging Speed Varies

Unlike filling a gas tank, EV charging speed is not constant. Multiple factors dynamically adjust how fast energy flows into the battery:

1. Battery State of Charge (SoC) Curve

DC fast charging follows a charging curve: - **0–20% SoC:** Charger can deliver near full rated power - **20–80% SoC:** Sustained high power (this is the “fast” part of fast charging) - **80–100% SoC:** Power tapers progressively to protect battery chemistry

This taper at high SoC is engineered behavior — it's not a fault. A driver who joins a charger at 85% will always see slow speeds.

Script: > “DC fast chargers automatically slow down once your battery gets above about 80% — that's a built-in protection for your battery's long-term health. It's completely normal and actually a good sign that your battery management system is working correctly.”

2. Thermal Management (Battery and Charger)

Both the vehicle battery and the charger's power modules generate heat during fast charging. If temperatures exceed safe thresholds: - The vehicle BMS will reduce the accepted charge rate - The charger's thermal protection will reduce output - In extreme cases, charging may pause briefly to allow cooling

This is most common in: - Hot ambient conditions (above 30°C / 86°F) - Cold ambient conditions (below 5°C / 41°F) — batteries also charge slowly when cold - Sessions following a previous DC fast charge (battery already warm) - Vehicles with less effective thermal management systems

Script: > “In hot weather, both your car's battery and the charger itself slow down to manage heat. It's a safety feature, not a malfunction — the speed should pick up if temperatures drop or if you wait a bit between charging sessions.”

3. Power Sharing Between Adjacent Vehicles

Many DC fast charging stations share a total power budget across multiple connector pairs. If two vehicles are charging simultaneously: - The total available power is divided between both sessions - The split may be equal or weighted (by vehicle capability, SoC, or arrival order, depending on firmware) - Adding a second vehicle to an adjacent port can immediately halve the power to both

Script: > “I can see there's another car charging at the adjacent port on that same unit. Some stations share power between vehicles plugged into the same unit. Moving to a different station — not just a different port on the same unit — would get you the full available speed.”

4. Vehicle Maximum Accepted Rate

Every EV has a maximum DC fast charging rate set by its hardware and BMS. Plugging a vehicle into a 350 kW charger doesn't mean the vehicle will accept 350 kW — it accepts only what it can handle:

Example vehicle	Max DC fast charge rate
Nissan LEAF (24–40 kWh)	50 kW
Nissan LEAF Plus (62 kWh)	100 kW
Hyundai IONIQ 5	230 kW
Kia EV6 GT	240 kW

Example vehicle	Max DC fast charge rate
Ford Mustang Mach-E	150 kW
Chevrolet Bolt EV	55 kW
Rivian R1T/R1S	220 kW

If the driver expects 350 kW but the vehicle can only accept 50 kW, the station is performing correctly.

Script: > “What’s the maximum fast charging speed for your vehicle? Some EVs are designed to accept a lower rate — the charger will always match what your car can handle, so the station speed rating is a ceiling, not a guarantee.”

5. Vehicle Charge Limit Set

Many drivers set a charge limit in their vehicle app or settings to extend battery longevity. A limit of 80% means the car will slow to nearly zero as it approaches 80% and stop entirely when reached.

Script: > “It’s worth checking your vehicle’s app or charging settings — a lot of drivers set a charge limit at 80 or 90 percent to protect their battery. If there’s a cap set, the charger naturally slows as you approach it.”

6. Grid Voltage Issues at the Site

If the electrical supply to the charging site has voltage irregularities or is undersized for the installed charger capacity, the charger’s power output will be limited. This is a genuine fault condition — neither the vehicle nor the charger firmware is throttling, but power input is insufficient.

Signs of a grid issue: - Consistently low output at the same station across multiple vehicles - Station shows “power limited” or a reduced output indicator - Output drops further when multiple chargers on site are in use simultaneously

This requires investigation by an electrician, not a remote reset.

Diagnostic Script — Slow Charging

Step 1 — Get the baseline numbers

“What speed are you currently seeing — does the screen or your vehicle’s display show a kW number or a miles-per-hour estimate?”

“And what speed were you expecting at this station?”

“What’s your current battery percentage?”

Step 2 — Apply the decision tree

Battery above 80%: → Expected taper. Explain SoC curve. No fault.

Cold or hot weather conditions: → Expected thermal management. Explain. No fault unless dramatically below expected (e.g., 10 kW from a 150 kW rated charger in mild weather).

Two vehicles at the same station unit: → Power sharing. Explain. Suggest moving to a different unit.

Vehicle's max charge rate is lower than the station rating: → Expected. Explain vehicle rate ceiling. No fault.

Driver has a charge limit set: → Expected. Explain limit. No fault — driver can adjust in vehicle settings.

None of the above apply AND the power is dramatically low: → Potential genuine fault. Proceed to Step 3.

Step 3 — Check backend power output

Check the station's reported output in the backend: - If reported output matches what the driver sees → the charger is reporting accurately; fault may be on vehicle side - If reported output is significantly higher than what the driver reports receiving → measurement discrepancy; log for investigation - If the charger shows "power limited" or a degraded output state in the backend → genuine charger fault

Step 4 — Soft reset for genuine fault

“What you're describing does sound like more than the usual throttling. I'm going to send a reset to the station — sometimes that restores the full output.”

Trigger soft reset. Ask driver to check the power level after the session restarts.

If power is restored: session was in a degraded state, likely a firmware or module issue that reset cleared. Log for monitoring.

If power is still dramatically low after reset: genuine hardware degradation or fault.

“The station still isn't delivering its full rated power. I'm going to flag it for a technician to inspect the power modules. I'm really sorry about this.”

Section A — Specific Slow-Charging Scenarios

Scenario: L2 Speed on a DCFC Port

Driver is at a DC fast charger but receiving Level 2 speeds (roughly 7–22 kW instead of 50–350 kW).

Most common causes: 1. **DC portion of CCS not connected properly** — only AC pins are making contact. Ask driver to unplug and re-seat firmly. 2. **Vehicle defaults to AC charging** — some vehicles require DC fast charge to be enabled in settings. Check vehicle documentation. 3. **Charger power module for DC has faulted** — the AC portion still works, DC portion is dead. This is a charger hardware fault. 4. **OCPP backend sent an AC-only session start** — a backend configuration error. Soft reset and retry.

Scenario: Charges Fast Initially Then Drops to Nearly Zero

Classic SoC taper at high state of charge. If this is happening below 80%, it may be: - Battery thermal derating (cell temperatures too high) - BMS conservatism (some manufacturers are more aggressive than others) - A degraded cell group in the battery pack causing BMS to limit overall charge rate

This is vehicle-specific and generally not actionable by charging network support.

“That slowdown near the top of the battery is normal for most EVs — the battery management system protects the cells by tapering the charge rate. If it’s dropping dramatically at a lower percentage than expected, it may be worth mentioning to your vehicle dealer, but the charger is performing correctly.”

Scenario: Power Fluctuates Up and Down During Session

Intermittent power that oscillates suggests: 1. Grid instability at the site (genuine electrical infrastructure issue — dispatch) 2. Vehicle BMS negotiating and renegotiating the charge rate (may be normal for some vehicles) 3. Connector partially seated — causing intermittent contact (ask driver to re-seat) 4. Thermal cycling — power dips briefly while cooling, then resumes (expected in hot conditions)

Scenario: Consistently Slow at This Station vs. Others

Driver reports same vehicle charges at full speed at other stations but always slow at this specific one:

This is almost certainly the charger’s power module operating at reduced capacity. Log the station ID, the reported speed, and the driver’s vehicle. Flag for a technician power output test — many network operators have a power output verification tool in the backend.

Escalation Criteria

Condition	Action
Power dramatically below station rating, no thermal/SoC explanation	Flag for tech inspection
L2 speeds at a DCFC port, re-seat and reset didn't help	Hardware fault dispatch
Same station slow for multiple different vehicles	Priority tech inspection
Driver reports power fluctuating wildly	Grid issue — escalate to tech (electrical)

Agent Reminders

- Never tell a driver the slow speed is “definitely normal” without walking through the decision tree — some genuine faults present like expected throttling.
- The SoC taper is the most common explanation — if the driver is at 80%+, lead with that.
- Cold batteries are a real limitation. Below 10°C / 50°F, DC fast charging can be significantly reduced — especially on vehicles without active battery preconditioning.
- Power sharing is often invisible to the driver — they don't see the other vehicle. Check the backend for simultaneous sessions on the same unit.
- If the charger shows full output but the vehicle reports slow charging, the issue is vehicle-side. Do not dispatch a tech to the charger.